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Project Progress Report

Week – 1 (22 Jan – 28 Jan) : Group Formation

In this week we formed a group of 4 members.

Members: Md. Faiaz Bin Hayder, Shahriar Swanon, Md. Abu Sufian Protik, Khandakar Atikur Rahman. We set up group chat and planned how to communicate and share files.

Week – 2 (29 Jan – 04 Feb) : Project Selection

This week we selected our project : **Create a custom image classification dataset featuring five classes: dog, cat, cow, lamb, zebra, each with 100 images sourced from the internet or captured using your phone. Develop a classification model to classify these classes with at least 90% accuracy.**

Week – 3 (05 Feb – 11 Feb) : Task Division

In this 3rd week, we discussed overall approach , tools and methodology. For this project we need to train model and that's why we need dataset. We divided the dataset collection among us , and I was assigned the Lamb class (100 Images).

Week – 4 (12 Feb – 18 Feb) : Lamb Dataset Collection

In this week I collected 100+ lamb images from the internet and phone camera. While choosing and taking the photos I ensured variety in the background , angle , and lighting across all images. Then, I organized and labelled images into the lamb class folder and shared with the group.

Week – 5 (19 Feb – 25 Feb) : NO Workings!!!!!!

In this week I was so sick , suffering from high fever, cough , that's why I couldn't make any progress for this project.

Week – 6 (26 Feb – 04 Mar) : Dataset Preparation & Feature Study

In the 6th week we combined all 5 class folders into one full 500+ image dataset. Studied about feature extraction : HOG for shape /edges , HSV color histograms, texture histogram. Learned about PCA and how it reduces feature dimensions while keeping important informations.

Week – 7 (05 Mar – 11 Mar) : Clear the ideas about training a model

In this week I tried to see some ML projects in you tube and other sites to ensure how model train actually works and how it can be done. Different types of project I observed ,some I can understand and some were so much difficult for me to understand.. but checking this problems thoroughly I am much clear about the model training part and how it can be done. Basically in this week I mainly focused to learn how model training can be done and how different types of model can be train with higher accuracy

Week – 8 (12 Mar – Today) : Trying to train a model

In this week I am trying to train a model about lamb class,,, I faces many difficulties while doing this,, specially dataset because images collected by different group members had different size formats and quality – had to resize and standardize everything before training . in this week I started SVM training , not fully done yet,.

